

Harshit Poddar

Stockholm, Sweden | harshit10233@gmail.com | [LinkedIn](#) | [Google Scholar](#) | [Portfolio](#) | [GitHub](#)

SUMMARY

Machine Learning Researcher specializing in **computer vision**, **multimodal systems**, and scalable **MLOps**. Dedicated to collaborating across teams to build robust pipelines and transition complex AI research into deployable solutions

EDUCATION

KTH Royal Institute of Technology || *Stockholm, Sweden*

2025 – Present

M.Sc. in Machine Learning

Vellore Institute of Technology || *Vellore, India*

2021 – 2025

B. Tech in Electronics & Communication Engineering (CGPA: 4.4/5)

SKILLS

- **Languages:** Python, MATLAB, SQL.
- **AI & ML:** PyTorch, YOLO, U-Net, LangChain (RAG), Hugging Face, Computer Vision, ChromaDB, Explainable AI (Grad-CAM, SHAP).
- **Cloud & MLOps:** Google Cloud Platform (GCP), MLflow, Git/GitHub, Linux.
- **Clinical Data & Standards:** DICOM, FHIR, OMOP.
- **Data & Engineering:** Agile Methodologies, System Validation, Power BI, Tableau, LaTeX.

WORK EXPERIENCE

Robotics Perception and Learning (RPL) Lab: *Student Researcher* || Stockholm, Sweden

Sep 2025 – Present

- Prototyping and optimizing a real-time, **multimodal AI navigation system** integrating **computer vision** and **audio processing**. Constructing dynamic **scene graphs** alongside researchers to enable spatial reasoning for autonomous agents. In collaboration with KTH AI Society.

AESIR (KTH Rocketry): *Systems Simulation Engineer* || Stockholm, Sweden

Sep 2025 – Present

- Developing the Freyja HIL (**Hardware-in-the-Loop**) system to simulate real-time aerospace sub-systems; performing rigorous Test Development & System Validation for embedded flight software to ensure mission reliability.

Defence Research & Development Organisation: *Summer Internship* || Bengaluru, India

Jun 2024 – Jul 2024

- Engineered an 8x60 phased array antenna and real-time GUI in **MATLAB**, achieving **25% efficiency gain** and **15% loss reduction** while performing radar degradation analysis for fault-tolerant system design.

TATA Memorial Hospital, ACTREC: *Research Internship* || Navi Mumbai, India

Nov 2023 – Dec 2023

- Deployed **U-Net** and **YOLOv8L** for multi-organ nuclei segmentation, balancing a **30% accuracy gain** against **20% faster** inference, and integrated **XAI methods (Grad-CAM/SHAP)** to validate the model's diagnostic reasoning.

Indian Space Research Organisation: *Research Internship* || Bengaluru, India

Sep 2023 – Oct 2023

- Led the optimization of a 16 x 16 Phased Antenna Array for MEOSAR satellite signals, achieving **>20% SNR improvement** through **Ansys HFSS** simulations and beamforming algorithms (**MVDR, LCMV**) for real-world orbit validation.

Indian Institute of Management: *Research Assistant* || Trichy, India

Mar 2023 – Jun 2023

- Developed Python scripts for **ETL (Extract, Transform, Load)** processes for 5,000+ audience responses and 200+ TV ads; applied NLP sentiment scoring and statistical modelling to quantify emotional impact on 10+ persuasion metrics.

TECHNICAL PROJECTS

AI-Driven Protocol Diagnostic Suite

Dec 2025 – Present

- Architecting a Generative AI pipeline using **LangChain** and **MLflow**, integrating **ChromaDB** for efficient vector storage and semantic retrieval of unstructured telemetry logs.

LungMate: AI-Driven Lung Cancer Diagnosis

Nov 2024 – Apr 2025

- Developed a hybrid Computer Vision system (EfficientNet-B3, Swin Transformer, CBAM, YOLOv11) for multi-metric tumor quantification from DICOM data, integrating a PyQt GUI for real-time enhancement and clinical baseline benchmarking

High-Altitude Autonomous Safety System (*Granted*)

Jun 2024 – Dec 2024

- Designed a real-time **Embedded System** for fall detection utilizing **Sensor Fusion** and automated actuator deployment for high-reliability safety applications.

Industrial IoT Safety Wearable (*Applied for Patent*)

Jan 2024 – May 2024

- Engineered a real-time ESP32 embedded system leveraging LoRaWAN and sensor fusion for critical safety telemetry in hazardous environments.

AWARDS, PUBLICATIONS & PATENTS

- **Patents & Research:** Filed 2 Patents (IoT Helmet, Fall Protection); **Published 15+ papers** in IEEE, Springer, and Elsevier.
- **Awards & AI:** 2× Raman Award; LLM Evaluation Participant at Google DeepMind & Andon Labs (Paid Honorarium).